

VANGUARDZ

GAME DESIGN DOCUMENT & TECHNICAL SPECIFICATIONS

Lead Game Designer: Kaizzcer

Development Status: Working Version 2.0 (Stable Release)

Systems Target: HTML5, React, Node.js, WebSocket Matrix

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1. Executive Summary & Story Matrix

Vanguardz is a tactical typing-action shooter that combines the speed and visual thrill of space combat with high-fidelity keyboard text inputs. Built as a cyberpunk spatial coordinate engine, players command fighter craft, clearing channels of letter-bullets, orbital debris, and hostiles. The game supports a robust solo progression and a real-time synchronized co-op multiplayer squadron module supporting up to three players.

1.1 The Vanguardz Narrative

"We watched our home world burn, a silent spark swallowed by the black ocean of space. The counselors told us that survival was enough, but to live on our knees is to fade into nothingness. Our friends carry the will to resist, but their ships are old, and their hearts are weary. If they engage the armada directly, they will be swept aside like space dust. So you made a choice. Under a shroud of stardust, you slipped into the dark alone, steering your fighter into the belly of the hostile empire. You do not seek glory—only to clear a path out of the shadows. We are the final shield. We will stand."

1.2 The Sacrifice Ending (Wave 100)

The narrative arc of Vanguardz reaches its resolution at **Wave 100**, the final gate. As the player enters the orbital sphere of the **Void Emperor**, they face a mathematically impossible onslaught of hostile fighters and a barrage of letter-bullets. The Void Emperor's shield matrix is locked, and completed word-nodes instantly regenerate. The battle is designed as a forced tactical sacrifice. Upon the destruction of the hull at Wave 100, the pilot does not receive a standard 'Ship Destroyed' failure message. Instead, they trigger the **Hero Sacrifice Ending**: the pilot's final explosion clears a warp path through the empire's main shield gates, allowing the rest of the Vanguardz fleet to bypass the blockades and secure a decisive victory.

2. Technical Architecture & Specifications

The Vanguardz platform is built using a modern JavaScript/Node stack optimized for high-performance visual loops and low-latency network synchronization.

Component	Technology	Description / Implementation
Core Logic & UI	React 19.2	Manages routing screens (Menu, Lobby, Docking, Play, GameOver) and player profiles.
Rendering & Math	HTML5 Canvas API	Implements the coordinate system, particle explosions, starfield, and laser rendering at 60fps.
Build Engine	Vite 8.1	Provides extremely fast hot-module reloading and optimized static asset chunking.
Styles & Theme	Vanilla CSS	Defines the HSL-based neon custom palettes, futuristic borders, and dark glass panels.
Co-op Backend	Node.js & Express 5.2	Hosts the game lobbies, registers pilot call signs, and coordinates socket rooms.
Lobby & Sync	WebSockets (ws 8.21)	Synchronizes enemy spawns, player positions, skill triggers, and health bars in real-time.
Audio Synthesis	Web Audio API	Procedurally synthesizes mechanical clicks, laser sweeps, and dynamic ingame melody tempos.

2.1 Architectural Flow Diagram

The game moves through a structured state loop:

- Profile Setup:** Persists pilot name and signature colors in LocalStorage.
- Lobby Sync:** Handshakes with WebSocket server to bind players by unique room codes.
- Docking Bay:** Triggered every 5 waves, allowing players to exchange weapons and unlock configurations.
- Game Loop:** Translates keyboard strokes into canvas coordinates, calculates particle collisions, and draws lasers.
- Ending Trigger:** At death, checks wave index. Wave 1-99 displays standard mission score; Wave 100+ triggers the final story sequence.

3. Detailed Hostile Enemy Specifications

To provide a complex typing experience, the game features **30 unique common enemy classes**. Each class contains a custom visual profile, specific flight mechanics, mathematical parameters, and targeting word lengths.

1. Nova Swarmer — *Drone*

Description	Small triangular vector ship that darts in rapid waves.
Behavioral Pattern	<i>Sinusoidal sideways wobble, speed scale: 1.2x</i>
Threat Index	Darts in groups of 3-4, focusing on lane corners.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	3-4 characters (e.g. KEY, SCAN, WARP)

2. Void Drifter — *Cruiser*

Description	Blocky heavy cargo ship that moves slowly down the screen.
Behavioral Pattern	<i>Straight vertical drift, speed scale: 0.5x</i>
Threat Index	Fires light single letter-bullets at player positions.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-7 characters (e.g. RADAR, MATRIX, HYBRID)

3. Neutron Interceptor — *Elite*

Description	Sharp arrow-shaped ship with glowing blue light rings.
Behavioral Pattern	<i>Diagonal zigzag pattern, speed scale: 1.0x</i>
Threat Index	Locked targets switch target indices rapidly on typos.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-6 characters (e.g. PLASMA, DRIFT, SECURE)

4. Quantum Replicator — *Special*

Description	Crystalline blue structure that oscillates periodically.
Behavioral Pattern	<i>Drift with periodic vertical stops, speed: 0.6x</i>
Threat Index	Splits into two smaller drones (Nova Swarmers) when half typed.
Combat Statistics	High (HP: 3)
Typing Vector Profile	6-8 characters (e.g. KINETIC, SYNAPSE, REPLICATE)

5. Solar Linker — *Support*

Description	Hexagonal gold support drone with energy nodes.
Behavioral Pattern	<i>Cosine weaving pattern, speed scale: 0.8x</i>
Threat Index	Creates a visible invulnerability beam to adjacent enemies.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	4-5 characters (e.g. LINK, SHIELD, NODE)

6. Chronos Raider — *Interceptor*

Description	Narrow hourglass-shaped stealth ship.
Behavioral Pattern	<i>Starts slow, accelerates by 5% every second</i>
Threat Index	Dives down rapidly if not targeted within 3 seconds of spawn.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	4-5 characters (e.g. CHRONO, RAPID, HASTE)

7. Nebula Stalker — *Special*

Description	Faint cloud-like vessel that blends into the background.
Behavioral Pattern	<i>Drift with periodic fading, speed: 0.7x</i>
Threat Index	Letters become completely invisible for 1.5 seconds every 3 seconds.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-7 characters (e.g. SHADOW, CLOAK, SPECTER)

8. Pulse Disruptor — *Jammer*

Description	Pulsating purple orb surrounded by magnetic rings.
Behavioral Pattern	<i>Static hover at middle height, speed: 0.4x</i>
Threat Index	Jams typing controls on typo, locking keystrokes for 1.2 seconds.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	6 characters (e.g. SIGNAL, SCRAMB, PULSAR)

9. Singularity Carrier — *Heavy*

Description	Massive carrier vessel with dark matter vents.
Behavioral Pattern	<i>Vertical drift, speed scale: 0.35x</i>
Threat Index	Spawns 3 small Helix Minions at its coordinate upon death.
Combat Statistics	High (HP: 3)
Typing Vector Profile	8-10 characters (e.g. GRAVITATIONAL, SINGULARITY)

10. Astra Kamikaze — *Special*

Description	Vibrant crimson needle-shaped fighter.
Behavioral Pattern	<i>Aggressive vertical dive target, speed scale: 1.8x</i>
Threat Index	Targets the player's coordinate and accelerates on typing mistakes.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	3-4 characters (e.g. RUSH, BURN, DIVE)

11. Spectral Cruiser — *Cruiser*

Description	Prismatic ship that shifts colors periodically.
Behavioral Pattern	<i>Weaving sinusoidal horizontal path, speed: 0.6x</i>
Threat Index	Changes its required typing color signature every 3 seconds.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-7 characters (e.g. PRISM, SHIFT, CHROMATIC)

12. Plasma Bomber — *Bomber*

Description	Bulky vector ship carrying glowing plasma bombs.
Behavioral Pattern	<i>Straight vertical drift, speed scale: 0.4x</i>
Threat Index	Drops drifting letter-mines that explode if they cross the bottom boundary.
Combat Statistics	High (HP: 3)
Typing Vector Profile	6-8 characters (e.g. WARHEAD, FISSION, BOMBARD)

13. Magnetar Pulsar — *Jammer*

Description	Bright yellow magnetar core with high rotation.
Behavioral Pattern	<i>Drifts left to right horizontally, speed: 0.5x</i>
Threat Index	Locks all equipped player skills for 5 seconds on spawn.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	7 characters (e.g. MAGNETIC, DRAIN, PULSE)

14. Cosmic Scavenger — *Support*

Description	Scrap-metal vessel with robotic claws.
Behavioral Pattern	<i>Weaves toward wreckage coordinates, speed: 0.8x</i>
Threat Index	Restores 50% health to itself when another enemy is destroyed.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-6 characters (e.g. RECYCLE, CLAW, SCRAP)

15. Warp Specter — *Special*

Description	Eerie glowing blue shape that shifts out of phase.
Behavioral Pattern	<i>Horizontal teleportation, speed scale: 0.9x</i>
Threat Index	Teleports randomly to another lane if the player commits a typo.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	5 characters (e.g. WARP, PHASE, GHOST)

16. Dark Matter Warden — *Defender*

Description	Heavy black barrier-shaped shield vessel.
Behavioral Pattern	<i>Moves in a block grid pattern, speed: 0.3x</i>
Threat Index	Possesses a physical shield that absorbs the first 3 keystrokes.
Combat Statistics	High (HP: 4)
Typing Vector Profile	7-9 characters (e.g. DEFENDER, FORTRESS, WARDEN)

17. Helix Minion — *Drone*

Description	Double-helix vector ship spinning constantly.
Behavioral Pattern	<i>Helical spiral downward path, speed scale: 1.1x</i>
Threat Index	Moves in circular loops, shifting the visual location of its text tag.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	4 characters (e.g. SPIN, TWIST, LOOP)

18. Titan Dreadnought — *Heavy*

Description	Massive flagship with multiple laser cannons.
Behavioral Pattern	<i>Slow vertical hover down, speed scale: 0.25x</i>
Threat Index	Requires three consecutive words to be typed to destroy the hull.
Combat Statistics	High (HP: 5)
Typing Vector Profile	8-12 characters (e.g. ANNIHILATION, DREADNOUGHT)

19. Supernova Anomaly — *Special*

Description	Flaming solar sphere that swells over time.
Behavioral Pattern	<i>Remains stationary in upper screen, speed: 0x</i>
Threat Index	Swells and triggers a full-screen screen shake explosion if not typed in 8s.
Combat Statistics	High (HP: 3)
Typing Vector Profile	6-7 characters (e.g. FUSION, REACTOR, NOVA)

20. Proton Charger — *Special*

Description	Twin-battery vector cell with blue spark lines.
Behavioral Pattern	<i>Drifts horizontally, speed scale: 0.7x</i>
Threat Index	Accumulates energy when typed, releasing a shield orb on destruction.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-6 characters (e.g. BATTERY, CHARGE, ENERGY)

21. Eclipse Shrouder — *Jammer*

Description	Umbral crescent ship that absorbs canvas light.
Behavioral Pattern	<i>Zigzag diagonal pattern, speed: 0.8x</i>
Threat Index	Blurs the player's health HUD and skill cooldowns while active.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	6-8 characters (e.g. ECLIPSE, OBSCURE, SHROUD)

22. Entropy Drainer — *Support*

Description	Spindle-shaped grey vessel that emits radiation.
Behavioral Pattern	<i>Weaves left to right slowly, speed: 0.5x</i>
Threat Index	Siphons the player's active score multiplier by 1 every 2 seconds.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	6 characters (e.g. DECAY, MATRIX, ERODE)

23. Vortex Sentinel — *Defender*

Description	Swirling vortex-shaped ship that attracts energy.
Behavioral Pattern	<i>Stays in center column, speed scale: 0.6x</i>
Threat Index	Applies a gravity pull, drawing nearby smaller ships into its lane.
Combat Statistics	Medium (HP: 3)
Typing Vector Profile	6-7 characters (e.g. VORTEX, GRAVITY, PULL)

24. Stardust Glider — *Drone*

Description	Graceful winged ship with a tail of particles.
Behavioral Pattern	<i>Glides horizontally before diving, speed: 1.2x</i>
Threat Index	Drops straight down after a short horizontal slide.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	4-5 characters (e.g. GLIDE, DUST, WIND)

25. Decoy Mirage — *Special*

Description	Prism-shaped ship that creates clones.
Behavioral Pattern	<i>Drifts straight down, speed scale: 0.75x</i>
Threat Index	Spawns 2 fake mirror clones that vanish in a single keystroke.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	5 characters (e.g. MIRAGE, CLONE, DECOY)

26. Ion Deflector — *Defender*

Description	Sleek mirror-plated shield ship.
Behavioral Pattern	<i>Weaves vertically and horizontally, speed: 0.7x</i>
Threat Index	Reflects standard laser beams, requiring a specific skill to damage.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	6 characters (e.g. REFLECT, DEFLECT, MIRROR)

27. Photon Striker — *Special*

Description	Gleaming golden fighter with solar sails.
Behavioral Pattern	<i>Straight vertical drift, speed scale: 1.1x</i>
Threat Index	Fires a rapid laser bullet at the player instantly if a typo is made.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	4-5 characters (e.g. FLASH, BEAM, LIGHT)

28. Apex Leviathan — *Heavy*

Description	Giant mechanical whale-shaped flagship.
Behavioral Pattern	<i>Slow vertical drift, speed scale: 0.2x</i>
Threat Index	Periodically deploys support shield linkers to protect other ships.
Combat Statistics	High (HP: 5)
Typing Vector Profile	9-11 characters (e.g. LEVIATHAN, BEHEMOTH)

29. Gamma Scavenger — *Support*

Description	Green-accented bio-mechanical collector.
Behavioral Pattern	<i>Drifts toward closest ship, speed: 0.9x</i>
Threat Index	Heals nearby ships for 25% health every 3 seconds of active flight.
Combat Statistics	Medium (HP: 2)
Typing Vector Profile	5-7 characters (e.g. TOXIC, CORRODE, SHIELD)

30. Void Empress Minion — *Drone*

Description	Ornate purple needle drone orbiting the core.
Behavioral Pattern	<i>Circular orbit around boss, speed scale: 1.3x</i>
Threat Index	Orbits the Wave 100 Boss, absorbing shots aimed at the main structure.
Combat Statistics	Low (HP: 1)
Typing Vector Profile	3-4 characters (e.g. GUARD, CORE, SHIELD)

4. Audio Synthesis & Procedural Soundtrack

Vanguardz features a custom, procedurally generated soundtrack and sound effects system built entirely on the HTML5 Web Audio API. This approach guarantees a zero-byte asset foot-print for core themes and enables the music to respond dynamically to gameplay states.

4.1 In-Game Generative Synth

The in-game music is generated using low-frequency oscillators (LFOs), sub-bass waves, and synthesizer tracks running in real-time. The tempo and complexity of the melody scales directly with the current wave number and threat level: as wave speed increases, the BPM of the synth drum track ramps up, creating physical tension in typing coordination.

4.2 Procedural SFX Mapping

Rather than relying on static audio files, SFX are synthesized on-demand with custom frequency sweeps:

- **Mechanical Key Clicks:** Generates a high-frequency metal contact leaf click (5500Hz bandpass noise) combined with a low frequency bottom thud (80Hz triangle wave) with random pitch variations (+/-10%) to simulate mechanical keyboard typing.
- **Laser Fire:** A rapid triangle wave frequency sweep starting at 1200Hz down to 400Hz over 120ms.
- **Shield Activations:** A sine wave sweep scaling exponentially from 180Hz up to 750Hz over 350ms, conveying energy charge.
- **Explosion SFX:** A combination of a low-pass filtered sawtooth wave (80Hz to 20Hz) and a triangle noise node generating rumbling impacts.

5. Game Rules, Controls & Skills Loadout

Vanguardz enforces typing accuracy as its core defensive shield. The rules are structured to balance typing speed, typo recovery, and skill deployment.

5.1 Core Gameplay Rules

- **Laser Locking:** Typing the first letter of a word locks the ship's firing system onto that specific target. The target remains locked until the word is completed or destroyed. In solo, all targets are orange/gray. In co-op, targets are color-coded (Red, Blue, Green) matching specific player spacecraft signatures. Players can only target enemies of their color (meteors are shared).
- **Typo Penalties:** Pressing an incorrect character triggers a screen shake, resets the player's active combo streak and multiplier to 1, and increases typing cooldowns.
- **Multiplier System:** Every 10 consecutive correct letters increases the score multiplier (up to 5x). Multiplier shield defense increases at higher multipliers (reducing hull damage on hits).

5.2 Equipped Skills Matrix

Every 5 waves, players enter the Orbital Docking Station, where they can choose to swap their active loadout (exactly 3 slots):

1. **EMP Discharge (Cooldown: 12s)**: Discharges a massive pulse that freezes all screen enemies and letter-bullets for 3.5 seconds.
2. **Overclock System (Cooldown: 15s)**: Doubles typing speed by auto-hitting two letters for every single key typed for 6 seconds.
3. **Nebula Veil (Cooldown: 18s)**: deploys a stellar cloud slowing all enemy movement speed by 50% for 8 seconds.
4. **Gravity Rewind (Cooldown: 20s)**: Reverses gravity locally, pushing all hostile ships back up the screen by 150 pixels.

5.3 Hardware Keyboard Controls

- **[A - Z]**: Type letters corresponding to highlighted hostile word-targets.
- **[1, 2, 3]**: Trigger equipped skill modules in slots 1, 2, and 3 respectively.
- **[Escape]**: Request pause / Resume game loop.
- **[Space]**: Confirm menus / Skip lobby cutscenes.